

because for an ideal current source $R_s = \infty$, so equivalent voltage source value will not be finite.

5.4.1 Source Transformation For Dependent Source

Source transformation is also applicable to dependent source in the same manner as for independent sources. It states that

An dependent voltage source V_x in series with a resistance R is equivalent to a dependent current source $I_x = V_x/R$, in parallel with a resistance R , keeping the controlling voltage or current unaffected.

or,

A dependent current source I_x in parallel with a resistance R is equivalent to an dependent voltage source $V_x = I_x R$, in series with a resistance R , keeping the controlling voltage or current unaffected.

Figure 5.4.4 shows the source transformation of an dependent source.



Fig. 5.4.4 Source Transformation of Dependent Sources

5.5 THEVENIN'S THEOREM

It states that any network composed of ideal voltage and current sources, and of linear resistors, may be represented by an equivalent circuit consisting of an ideal voltage source, V_{Th} , in series with an equivalent resistance, R_{Th} as illustrated in the figure 5.5.1.

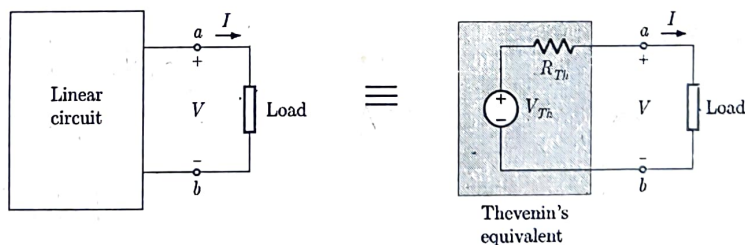


Fig. 5.5.1 Illustration of Thevenin Theorem

where V_{Th} is called Thevenin's equivalent voltage or simply Thevenin voltage and R_{Th} is called Thevenin's equivalent resistance or simply Thevenin resistance.

The methods of obtaining Thevenin equivalent voltage and resistance are given in the following sections.

5.5.1 Thevenin's Voltage

$R_{Th} = (\text{Thevenin's Resistance})$ Page 215
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The equivalent Thevenin voltage (V_{Th}) is equal to the open-circuit voltage present at the load terminals (with the load removed). Therefore, it is also denoted by V_{oc} .

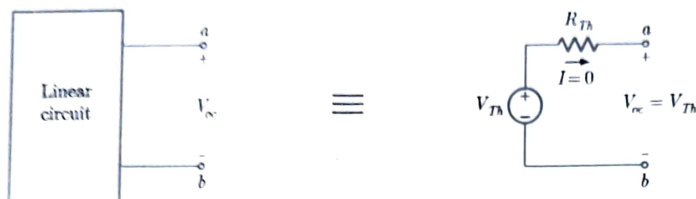


Fig. 5.5.2 Equivalence of Open circuit and Thevenin Voltage

Figure 5.5.2 illustrates that the open-circuit voltage, V_{oc} , and the Thevenin voltage, V_{Th} , must be the same because in the circuit consisting of V_{Th} and R_{Th} , the voltage V_{oc} must equal V_{Th} , since no current flows through R_{Th} and therefore the voltage across R_{Th} is zero. Kirchhoff's voltage law confirms that

$$V_{Th} = R_{Th}(0) + V_{oc} = V_{oc}$$

The procedure of obtaining Thevenin voltage is given in the following methodology.

M E T H O D O L O G Y 1

1. Remove the load i.e. open circuit the load terminals.
2. Define the open-circuit voltage V_{oc} across the open load terminals.
3. Apply any preferred method (KCL, KVL, nodal analysis, mesh analysis etc.) to solve for V_{oc} .
4. The Thevenin voltage is $V_{Th} = V_{oc}$.

NOTE :

Note that this methodology is applicable with the circuits containing both the dependent and independent source.

If a circuit contains dependent sources only, i.e. there is no independent source present in the network then its open circuit voltage or Thevenin voltage will simply be zero.

NOTE :

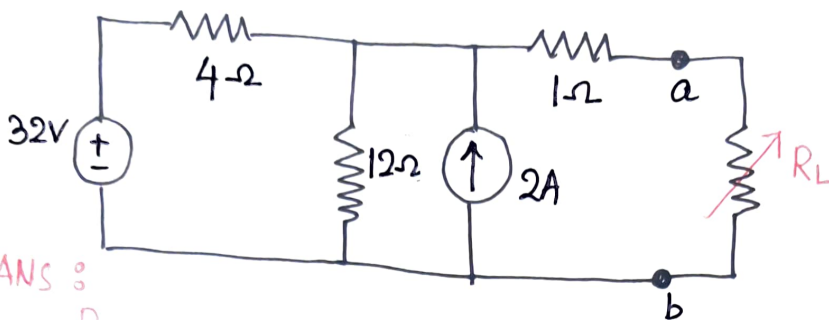
For the Thevenin voltage we may use the terms Thevenin voltage or open circuit voltage interchangeably.

5.5.2 Thevenin's Resistance

Thevenin resistance is the input or equivalent resistance at the open circuit terminals a, b when all independent sources are set to zero (voltage sources replaced by short circuits and current sources replaced by open circuits).

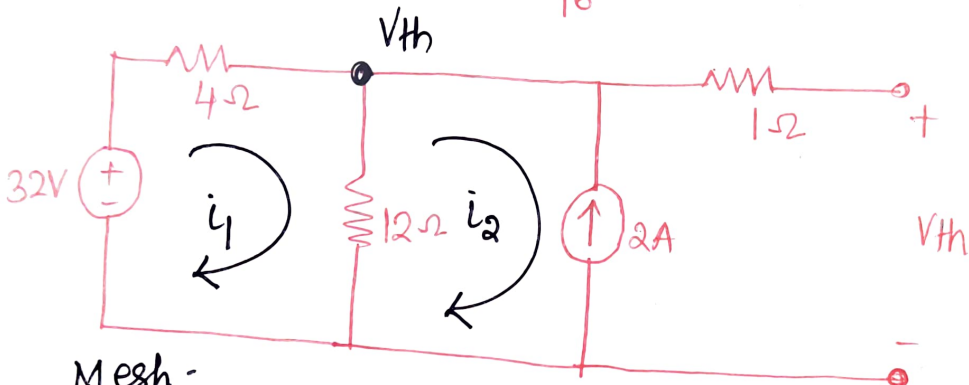
We consider the following cases where Thevenin resistance R_{Th} is to be determined.

Q. Find the Thevenin's Equivalent ckt for the circuit shown. Then find the current through $R_L = 6\Omega$, 16Ω and 36Ω



ANS :

$$R_{th} = 4 \parallel 12 + 1 = \frac{12 \times 4}{16} + 1 = 4\Omega$$



Mesh -

$$\Rightarrow 16i_1 - 12i_2 = 32$$

$$\Rightarrow i_2 = -2A$$

$$16i_1 + 24 = 32$$

$$i_1 = \frac{32 - 24}{16} = 0.5A$$

$$\begin{aligned} V_{th} &= 12(i_1 - i_2) \\ &= 12(0.5 - (-2)) \\ &= 30V \end{aligned}$$

Nodal

$$\frac{V_{th} - 32}{4} + \frac{V_{th}}{12} = 2$$

$$\frac{3V_{th} - 96 + V_{th}}{12} = 2$$

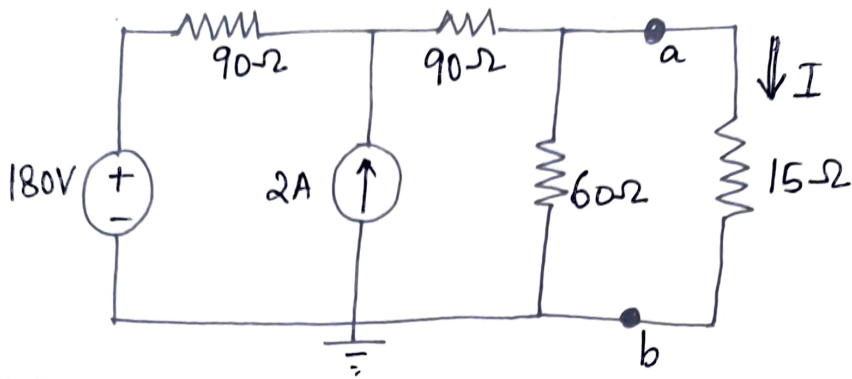
$$4V_{th} = 24 + 96$$

$$\begin{aligned} V_{th} &= \frac{120}{4} \\ &= 30V \end{aligned}$$

$$\text{So, } I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{30}{4 + R_L}$$

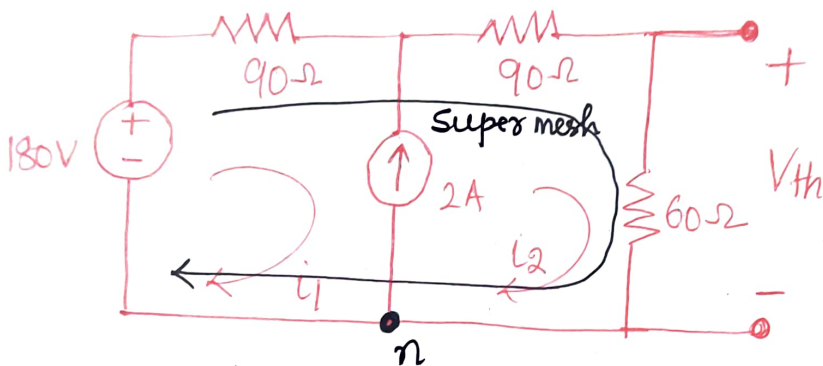
$$\Rightarrow R_L = 6 \quad I_L = 3A \quad \Rightarrow R_L = 16 \quad I_L = 1.5A \quad \Rightarrow R_L = 36 \quad I_L = 0.75A$$

Q. Using Thevenin's Theorem find the equivalent CKT and then find I ?



ANS

$$R_{th} = \frac{180 \times 60}{240} = 45\Omega$$



At node 'n'

$$i_2 = 2 + i_1$$

Supermesh eqn $\rightarrow$

$$-180 + 90i_1 + 90i_2 + 60i_2 = 0$$

$$-180 + 90i_1 + 150(2 + i_1) = 0$$

$$-180 + 90i_1 + 300 + 150i_1 = 0$$

$$240i_1 = -300 + 180$$

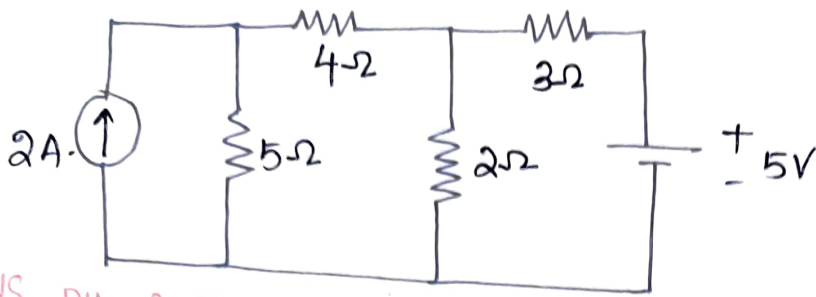
$$i_1 = \frac{-120}{240} = -0.5A$$

$$i_2 = 2 - 0.5 = 1.5A$$

$$V_{th} = i_2 \times 60\Omega = 1.5 \times 60 = 90V$$

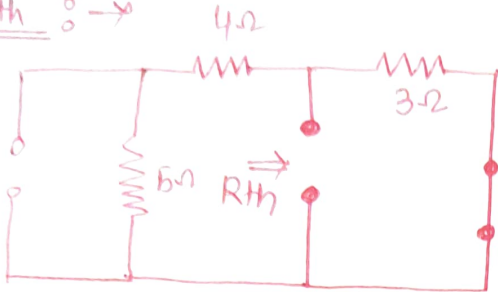
$$I_L = \frac{90}{45 + 15} = \underline{\underline{1.5A}}$$

Q. Find the current flowing in the 2Ω resistor of the ckt shown by Thevenin's Theorem :



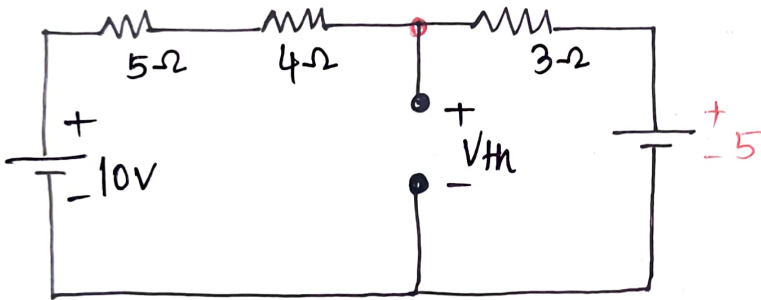
ANS

R_{th} : $\rightarrow$



$$R_{th} = \frac{9 \times 3}{9 + 3} = \frac{27}{12} = 2.25\Omega$$

V_{th} : $\rightarrow$

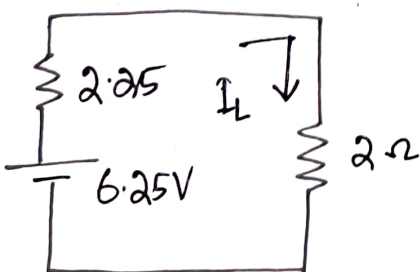


$$\Rightarrow \frac{V_{th} - 10}{9} + \frac{V_{th} - 5}{3} = 0$$

$$\Rightarrow \frac{V_{th} - 10 + 3V_{th} - 15}{9} = 0$$

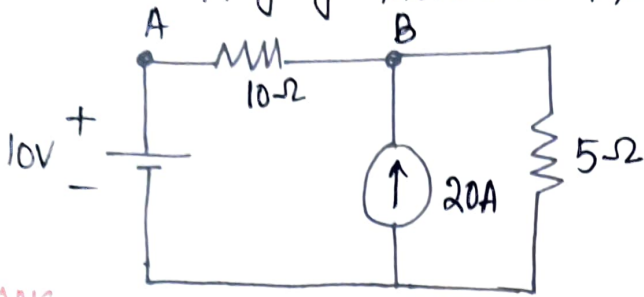
$$\Rightarrow 4V_{th} = 25$$

$$\Rightarrow V_{th} = \frac{25}{4} = 6.25V$$



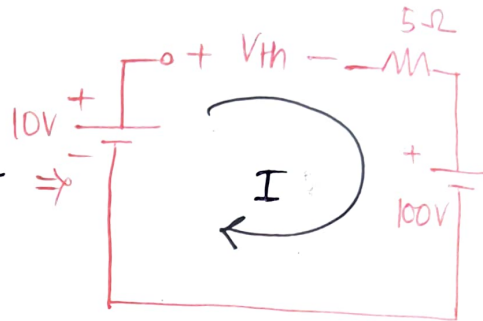
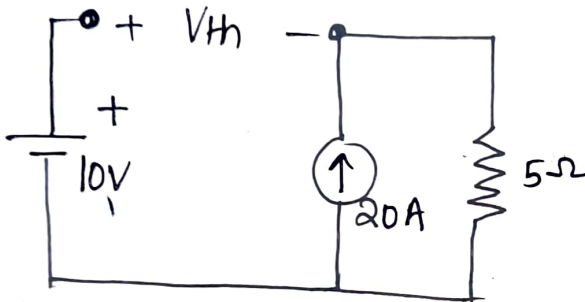
$$I_L = \frac{6.25}{2.25 + 2} = 1.47A$$

Q. Find the current flowing in the 10Ω resistor in the ckt applying Thevenin's Theorem $\rightarrow$



ANS $R_{th} = 5\Omega$

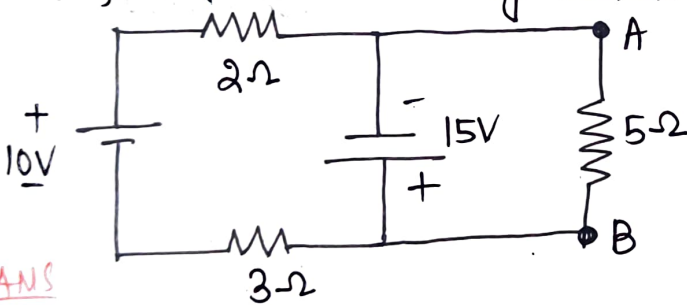
$V_{th} = ?$



$$V_{th} = V_{AB} = -10 + 100 = 90V$$

$$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{90}{15} = 6A$$

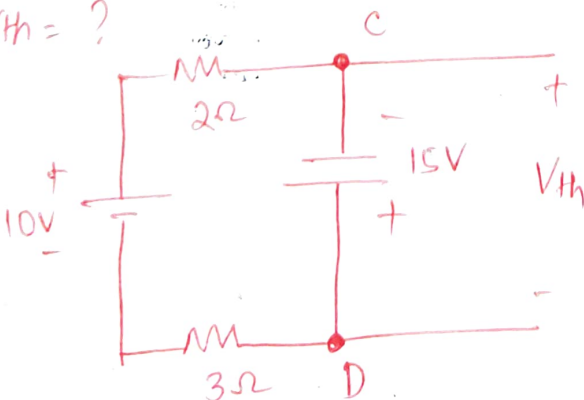
Q. Find the current flowing through 5Ω resistor of the circuit using Thevenin's Theorem $\rightarrow$



ANS

$R_{th} = 0$

$V_{th} = ?$



$$V_{th} = V_{AB} = V_{CD}$$

$$\Rightarrow \boxed{V_{th} = -15V}$$

$$I_L = \frac{V_{th}}{R_L + R_{th}}$$

$$= \frac{-15}{5} = -3A$$

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circuit. A large circuit may be replaced by a single independent voltage source and a single resistor. The equivalent network behaves the same way externally as the original circuit. Consider a linear circuit terminated by a load R_L , as shown in figure 5.5.5. The current I_L through the load and the voltage V_L across the load are easily determined once the Thevenin equivalent of the circuit at the load's terminals is obtained.

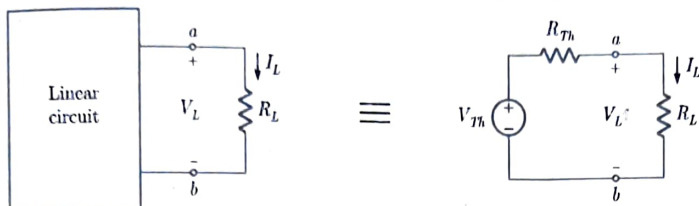


Fig. 5.5.5 A Circuit with a Load and its Equivalent Thevenin Circuit

Current through the load R_L

$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

Voltage across the load R_L

$$V_L = R_L I_L = \frac{R_L}{R_{Th} + R_L} V_{Th}$$

5.6 NORTON'S THEOREM

Any network composed of ideal voltage and current sources, and of linear resistors, may be represented by an equivalent circuit consisting of an ideal current source, I_N , in parallel with an equivalent resistance, R_N as illustrated in figure 5.6.1.

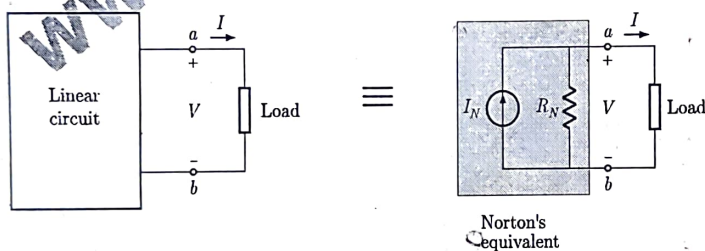


Fig. 5.6.1 Illustration of Norton Theorem

where I_N is called Norton's equivalent current or simply Norton current and R_N is called Norton's equivalent resistance. The methods of obtaining Norton equivalent current and resistance are given in the following sections.

5.6.1 Norton's Current

The Norton equivalent current is equal to the short-circuit current that would flow when the load replaced by a short circuit. Therefore, it is also called short circuit current I_{sc} .

NORTON'S Theorem :
It states that a linear two-terminal ckt can be replaced by an equivalent ext consisting of a current source I_N in parallel with resistor (R_N). ✓

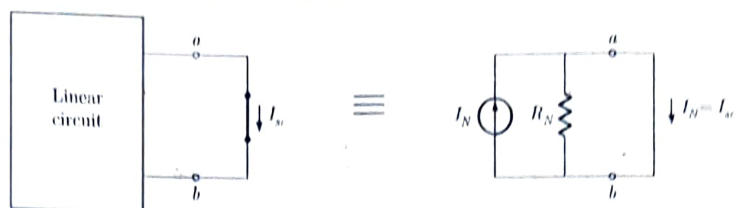


Fig 5.6.2 Equivalence of Short Circuit Current and Norton Current

Figure 5.6.2 illustrates that if we replace the load by a short circuit, then current flowing through this short circuit will be same as Norton current I_N

$$I_N = I_{sc}$$

The procedure of obtaining Norton current is given in the following methodology. Note that this methodology is applicable with the circuits containing both the dependent and independent source.

M E T H O D O L O G Y

1. Replace the load with a short circuit.
2. Define the short circuit current, I_{sc} , through load terminal.
3. Obtain I_{sc} using any method (KCL, KVL, nodal analysis, loop analysis).
4. The Norton current is $I_N = I_{sc}$.

If a circuit contains dependent sources only, i.e. there is no independent source present in the network then the short circuit current or Norton current will simply be zero.

5.6.2 Norton's Resistance

Norton resistance is the input or equivalent resistance seen at the load terminals when all independent sources are set to zero (voltage sources replaced by short circuits and current sources replaced by open circuits) i.e. Norton resistance is same as Thevenin's resistance

$$R_N = R_{Th}$$

So, we can obtain Norton resistance using same methodologies as for Thevenin resistance. Dependent and independent sources are treated the same way as in Thevenin's theorem.

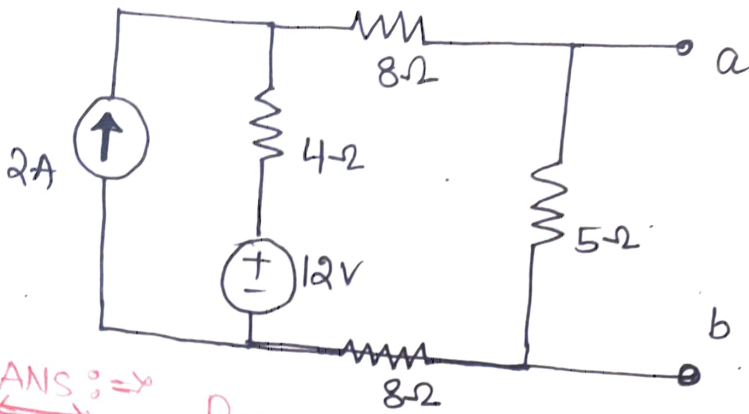
NOTE :

For the Norton current we may use the term Norton current or short circuit current interchangeably.

5.6.3 Circuit Analysis Using Norton's Equivalent

As discussed for Thevenin's theorem, Norton equivalent is also useful in circuit analysis. It simplifies a circuit. Consider a linear circuit terminated by a load R_L , as shown in figure 5.6.4. The current I_L through the load and the voltage V_L across the load are easily determined once the Norton equivalent of the circuit at the load's terminals is obtained,

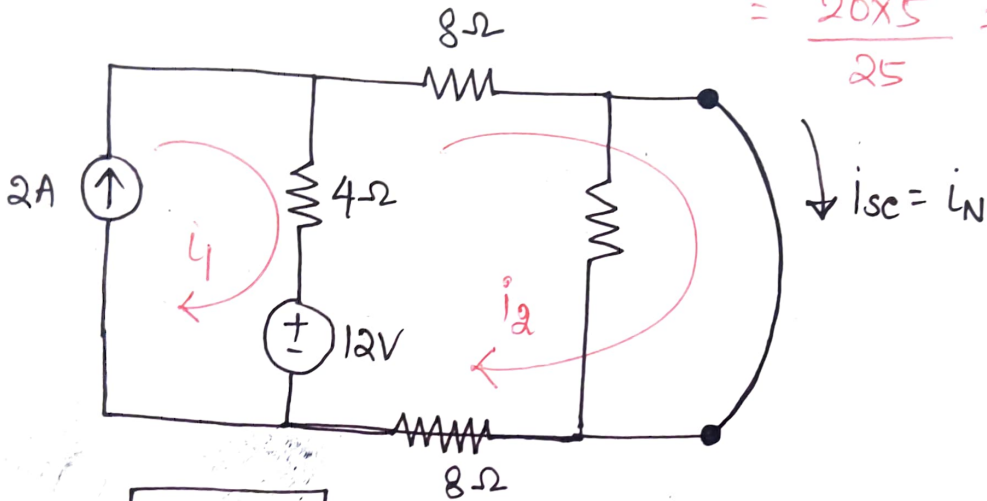
Q. Find the Norton equivalent ckt at terminals (a-b)



ANS: $\Rightarrow$

$$R_N = 5 \parallel (8 + 8 + 4) = 5 \parallel 20$$

$$= \frac{20 \times 5}{25} = 4\Omega$$

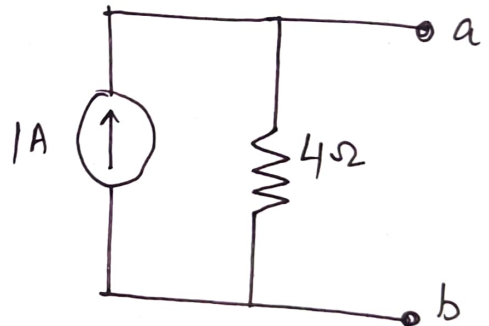


M-1 $i_1 = 2A$

M-2 $20i_2 - 4i_1 = 12$

$i_2 = 1A$

$i_2 = I_{sc} = I_N = 1A$



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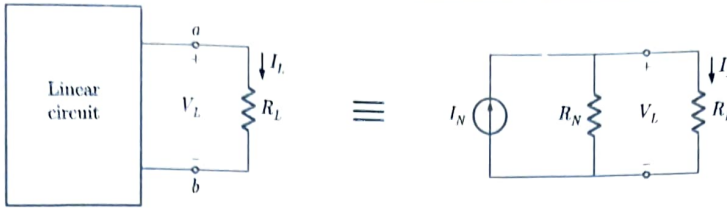


Fig. 5.6.4 A circuit with a Load and its Equivalent Norton Circuit

Current through load R_L is,

$$I_L = \frac{R_N}{R_L + R_N} I_N$$

Voltage across load R_L is,

$$V_L = R_L I_L = \frac{R_L R_N}{R_N + R_L} I_N$$

5.7 TRANSFORMATION BETWEEN THEVENIN & NORTON'S EQUIVALENT CIRCUITS

From source transformation it is easy to find Norton's and Thevenin's equivalent circuit from one form to another as following



Fig. 5.7 Source Transformation of Thevenin and Norton Equivalents

5.8 MAXIMUM POWER TRANSFER THEOREM

Maximum power transfer theorem states that a load resistance R_L will receive maximum power from a circuit when the load resistance is equal to Thevenin's/ Norton's resistance seen at load terminals.

i.e. $R_L = R_{Th}$, (For maximum power transfer)

In other words a network delivers maximum power to a load resistance R_L when R_L is equal to Thevenin equivalent resistance of the network.

PROOF :

Consider the Thevenin equivalent circuit of figure 5.8.1 with Thevenin voltage V_{Th} and Thevenin resistance R_{Th} .

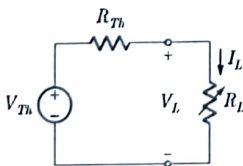


Fig. 5.8.1 A Circuit Used for Maximum Power Transfer

We assume that we can adjust the load resistance R_L . The power absorbed by the load, P_L , is given by the expression

$$P_L = I_L^2 R_L \quad (5.8.1)$$

and that the load current is given as,

$$I_L = \frac{V_{Th}}{R_L + R_{Th}} \quad (5.8.2)$$

Substituting I_L from equation (5.8.2) into equation (5.8.1)

$$P_L = \frac{V_{Th}^2}{(R_L + R_{Th})^2} R_L \quad (5.8.3)$$

To find the value of R_L that maximizes the expression for P_L (assuming that V_{Th} and R_{Th} are fixed), we write

$$\frac{dP_L}{dR_L} = 0$$

Computing the derivative, we obtain the following expression :

$$\frac{dP_L}{dR_L} = \frac{V_{Th}^2 (R_L + R_{Th})^2 - 2V_{Th}^2 R_L (R_L + R_{Th})}{(R_L + R_{Th})^4}$$

which leads to the expression

$$(R_L + R_{Th})^2 - 2R_L (R_L + R_{Th}) = 0$$

or

$$R_L = R_{Th}$$

Thus, in order to transfer maximum power to a load, the equivalent source and load resistances must be matched, that is, equal to each other.

$$R_L = R_{Th}$$

The maximum power transferred is obtained by substituting $R_L = R_{Th}$ into equation (5.8.3)

$$P_{max} = \frac{V_{Th}^2 R_{Th}}{(R_{Th} + R_{Th})^2} = \frac{V_{Th}^2}{4R_{Th}} \quad (5.8.4)$$

or,

$$P_{max} = \frac{V_{Th}^2}{4R_L}$$

If the Load resistance R_L is fixed :

Now consider a problem where the load resistance R_L is fixed and Thevenin resistance or source resistance R_s is being varied, then

$$P_L = \frac{V_{Th}^2}{(R_L + R_s)^2} R_L$$

To obtain maximum P_L denominator should be minimum or $R_s = 0$. This can be solved by differentiating the expression for the load power, P_L , with respect to R_s instead of R_L .

The step-by-step methodology to solve problems based on maximum power transfer is given as following :

M E T H O D O L O G Y

1. Remove the load R_L and find the Thevenin equivalent voltage V_{Th} and resistance R_{Th} for the remainder of the circuit.
2. Select $R_L = R_{Th}$, for maximum power transfer.
3. The maximum average power transfer can be calculated using $P_{max} = V_{Th}^2 / 4R_{Th}$.